6CS012 – Artificial Intelligence and Machine Learning.
Tutorial – 03
Perceptron Learning Algorithm
AND
Realization of XOR Problem.

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1. Understanding the Neurons.

{ Foundational Building Block of Artificial Neural Network and Inspired by Human Brain.}

1.1 MuCulloch – Pits Neurons.

- aka MCP neuron was a first computational model of a human neuron proposed by Warren MuCulloch and Walter Pitts in 1943.
- Following is the Mathematical Formalization of MCP:
  - MCP describes the activity of single neuron with **two states:** firing(1) or not firing(0).

For input $x_i \in \{0, \text{or } 1\}$ and output $y_i = \{0 \text{ or } 1\}$

{

$y = 0$ if any x_i is inhibitory,

else:

{

$g(x_1, x_2, x_3 \ldots, x_n) = g(X) = \sum_{i=1}^{n} x_i$

{

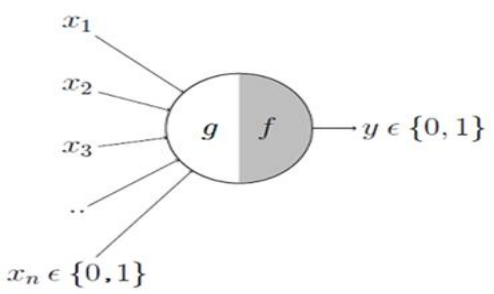
if: $g(X) \geq T$

$y = f(g(X)) = 1$

else: $g(X) < T$

$y = f(g(X)) = 0$

}



We will use this notation for easy representations.

1.2 Limitations of MCP Neurons.

- A single **McCulloch Pitts Neuron** can be used to represent boolean functions which are linearly separable.
  - **Linear separability** (**for boolean functions**) :
    - There exists a line (plane) such that
      - all inputs which produce a 1 lie on one side of the line (plane)
      - and all inputs which produce a 0 lie on other side of the line (plane)
- The **MCP Neuron Architecture** lacks **several characteristics** of **biological networks**:
  - **Complex connectivity patterns i.e. represents single neuron only**
  - **Processing of continuous values**
  - **A measure of importance**
  - **A learning procedure**
- Regardless of limitation, MCP is considered **a significant** first step towards development of ANN models

- **Towards Better Model…**

1.3 The Perceptron.

- The **Perceptron**, proposed by **Rosenblatt(1958),**
  - was an improvement over **McCulloch-Pitts (MCP) neurons**,
    - introducing a computational model with
      - **weighted inputs** and a **decision function** along with **learning theory** for **classification task**.

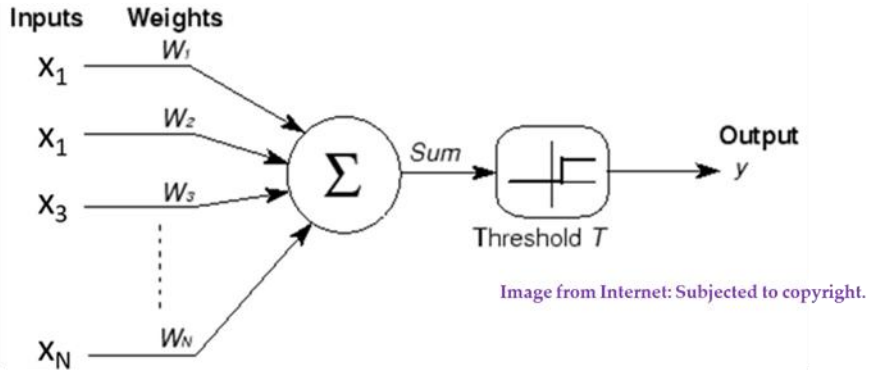


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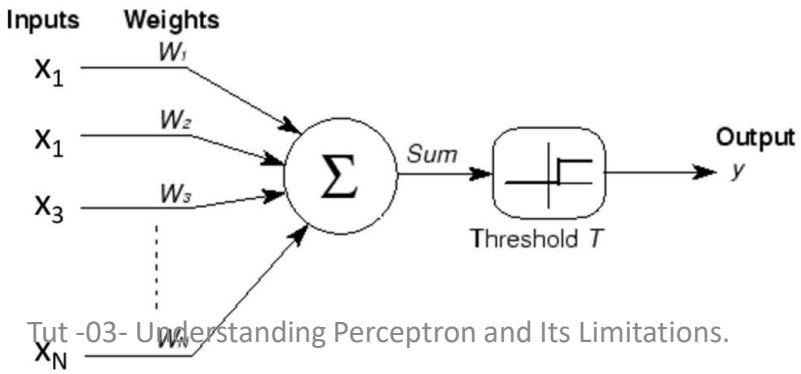
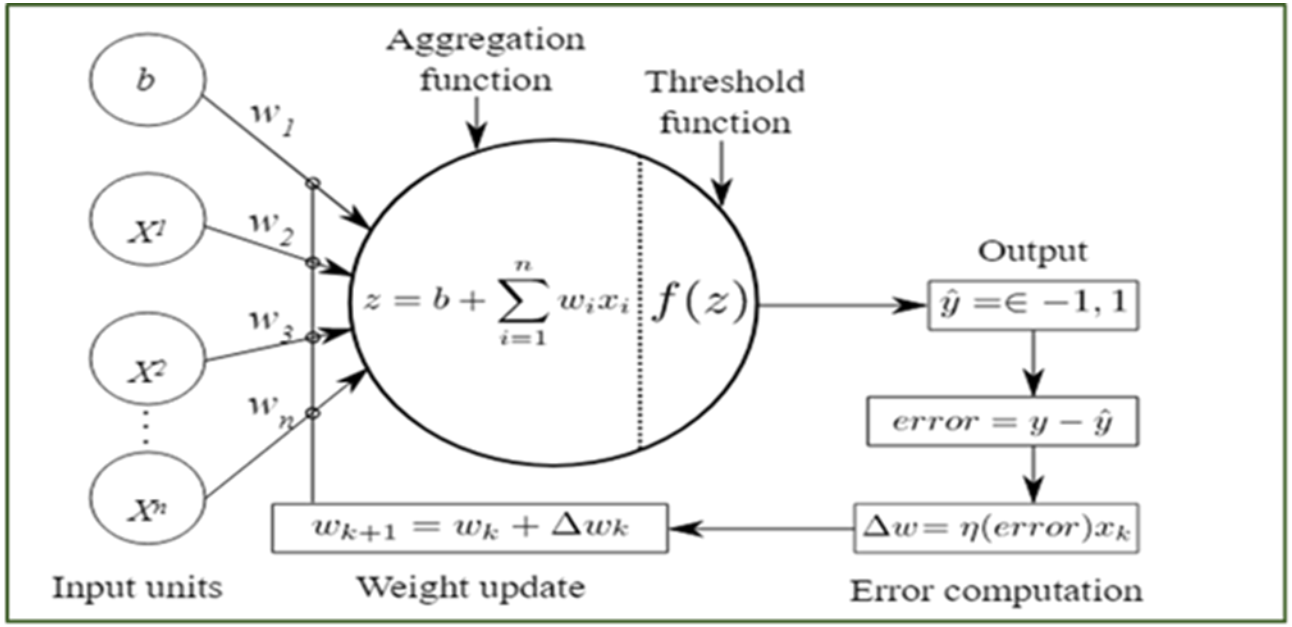
Fig: A Computational Representation of The Perceptron.

1.4 Simplified Computational Representation of Perceptron.

- **Mathematical Formulation:**
  - A perceptron takes a set of inputs $\mathbf{x_1, x_2, \ldots, x_n}$, assigns them weights $\mathbf{w_1, w_2, \ldots, w_n}$ and computes a weighted sum:
    - $\mathbf{z = \sum_{i=1}^{n} w_i x_i + w_o}$
    - here:
      - $w_i \rightarrow$ weights learned during training.
      - $w_o \rightarrow$ bias also learned during training. (adjusts the decision boundary)
      - $z \rightarrow$ net weighted input.
- The perceptron model then applies a threshold activation function (aka step function) on **z (net weighted input)** given by:
  - $\mathbf{f(z) = \begin{cases} 1 \text{ if } z \geq 0 \\ 0 \text{ if } z < 0 \end{cases}}$
  - This **threshold function** determines whether the perceptron activates (outputs 1) or remains inactive (outputs 0).



1.4.1 The Perceptron Learning Algorithm.



The Perceptron with Learning Algorithm

Fig: A Pictorial Representation of Perceptron Learning Algorithm.

1.4.2 Implementation of Perceptron Learning Algorithm.

Perceptron Learning Algorithm with Random Initialization

Algorithm 1 Perceptron Learning Algorithm

Require: Training dataset $D = \{(x_i, y_i)\}$, where:
x_i is the input feature vector (including bias term)
$y_i \in \{0, 1\}$ or $y_i \in \{-1, 1\}$
Require: Learning rate η (small positive value, e.g., 0.01)
Require: Number of epochs (max iterations)
Initialize: Randomly assign small values to $w = (w_1, w_2, \ldots, w_n)$
Randomly initialize bias b (or include it in w)
for each epoch **do**
  ConvergenceFlag = **True**
  for each training sample (x_i, y_i) **do**
    Compute weighted sum:

$$z = \sum w_i x_i + b$$

    Apply activation function (step function):

$$\hat{y} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$$

    if $\hat{y} \neq y$ **then** ▷ Update weights if misclassified
      Update weights:

$$w_i = w_i + \eta(y - \hat{y})x_i$$

      Update bias:

$$b = b + \eta(y - \hat{y})$$

      ConvergenceFlag = **False**
    end if
  end for
  if ConvergenceFlag is **True then**
    Break ▷ Stop if no updates occur (convergence)
  end if
end for
Output: Learned weights w and bias b

2. The Perceptron and Boolean Function Approximations.

{ Can we use Perceptron Learning Theory to Learn Boolean function?}

2.1 Perceptron for "OR" Function.

OR Function

Step 1: Define the OR Function

The OR function follows this truth table:

x_1	x_2	OR Output y
0	0	0
0	1	1
1	0	1
1	1	1

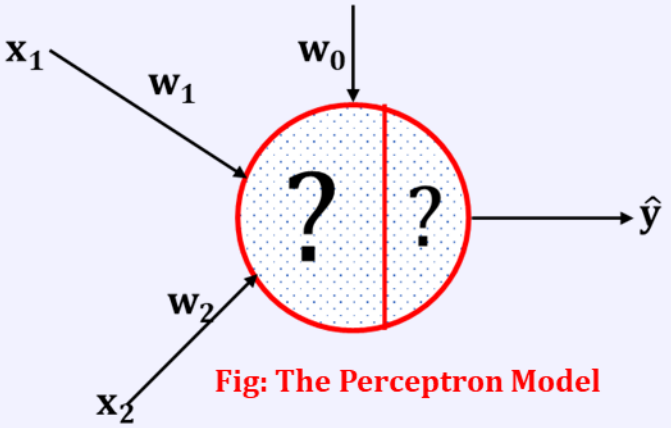


Fig: The Perceptron Model

The perceptron will learn the weights (w_1, w_2) and bias w_0 to correctly classify these examples.

3.1.1 Step – 2: Initialization of Weights.

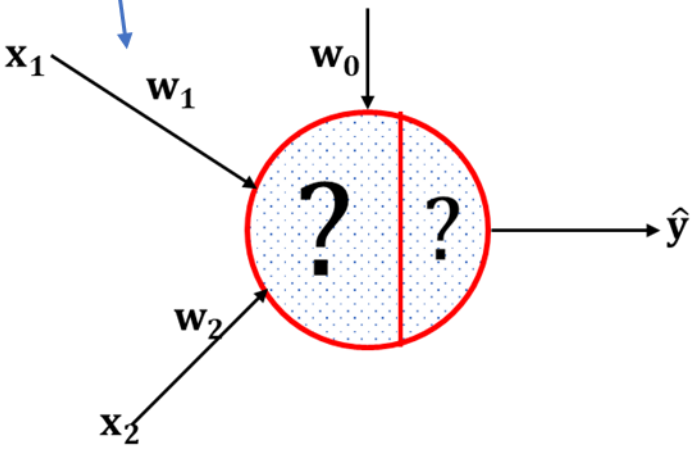
Step 2: Initialize Weights Randomly

We initialize weights with small random values:

$$w_1 = 0.2, \quad w_2 = -0.1, \quad w_0 = 0.1 \text{ (bias)}$$

Learning rate:

$$\eta = 0.1$$



3.1.2 Step – 3: Start of the Training Process.

Step 3: Training Process (Epoch 1)

For each training example, compute the weighted sum:

$$z = w_1 x_1 + w_2 x_2 + w_0$$

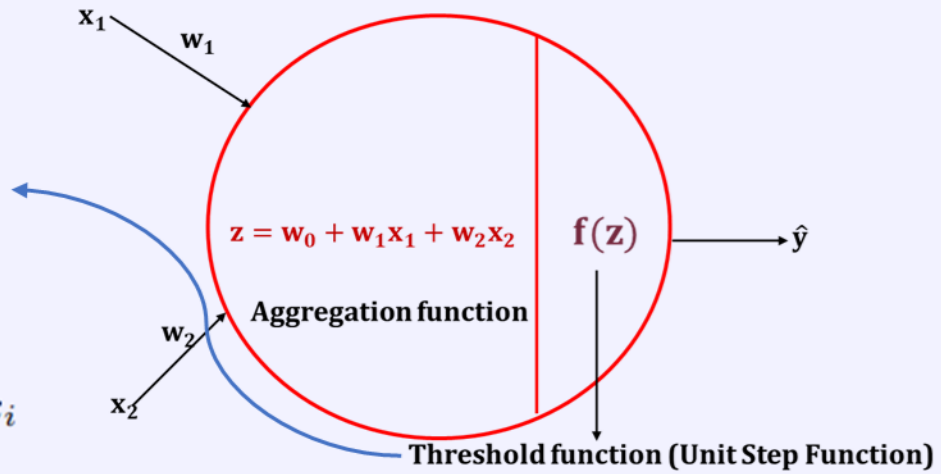
Apply the step function:

$$\hat{y} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$$

Update rule (if $y \neq \hat{y}$):

$$w_i = w_i + \eta(y - \hat{y})x_i$$

$$w_0 = w_0 + \eta(y - \hat{y})$$



✓ Let's Start Learning .....

3.1.3 Step – 4: Epoch – 1 – Step by Step Weight update.

1st Training Example: (0,0) $\rightarrow$ Expected $y = 0$

Compute weighted sum:

$$z = (0.2 \times 0) + (-0.1 \times 0) + 0.1 = 0.1$$

Apply step function:

$$\hat{y} = 1 \quad (\text{since } 0.1 \geq 0)$$

Misclassified ($y \neq \hat{y}$), so update weights:

$$w_1 = 0.2 + 0.1(0 - 1) \times 0 = 0.2$$

$$w_2 = -0.1 + 0.1(0 - 1) \times 0 = -0.1$$

$$w_0 = 0.1 + 0.1(0 - 1) = 0.0$$

3.1.3 Step – 4: Epoch – 1 – Step by Step Weight update.

2nd Training Example: (0,1) → Expected $y = 1$

Compute weighted sum:

$$z = (0.2 \times 0) + (-0.1 \times 1) + 0.0 = -0.1$$

Apply step function:

$$\hat{y} = 0 \quad (\text{since } -0.1 < 0)$$

Misclassified, so update weights:

$$w_1 = 0.2 + 0.1(1 - 0) \times 0 = 0.2$$

$$w_2 = -0.1 + 0.1(1 - 0) \times 1 = 0.0$$

$$w_0 = 0.0 + 0.1(1 - 0) = 0.1$$

3.1.3 Step – 4: Epoch – 1 – Step by Step Weight update.

3rd Training Example: (1,0) → Expected $y = 1$
Compute weighted sum:

$$z = (0.2 \times 1) + (0.0 \times 0) + 0.1 = 0.3$$

Apply step function:

$$\hat{y} = 1 \quad (\text{since } 0.3 \geq 0)$$

Correctly classified ($y = \hat{y}$), so no update.
4th Training Example: (1,1) → Expected $y = 1$
Compute weighted sum:

$$z = (0.2 \times 1) + (0.0 \times 1) + 0.1 = 0.3$$

Apply step function:

$$\hat{y} = 1 \quad (\text{since } 0.3 \geq 0)$$

Correctly classified, so no update.

3.1.4 Epoch – 1 – Final Weight Update.

Weight Updates During Epoch 1

Training Sample (x_1, x_2, y)	**Computed** z	**Predicted** $\hat{y}$	**Error** $y - \hat{y}$	**Weight Updates**
$(0, 0, 0)$	0.1	1	$0 - 1 = -1$	$w_0 = 0.1 - 0.1 = 0.0$
$(0, 1, 1)$	-0.1	0	$1 - 0 = 1$	$w_2 = -0.1 + 0.1 = 0.0$, $w_0 = 0.0 + 0.1 = 0.1$
$(1, 0, 1)$	0.3	1	$1 - 1 = 0$	No update
$(1, 1, 1)$	0.3	1	$1 - 1 = 0$	No update

- **Final Weights at the end of Epoch 1:**
  1. $w_1 = 0.2 \rightarrow$ unchanged.
  2. $w_2 = 0.0$
  3. $w_0 = 0.1$

3.1.5 Start of Epoch – 2.

$\mathbf{z = w_0 + w_1x_1 + w_2x_2}$
From Epoch 1:
$\mathbf{z = 0.1 + 0.2 \times 0 + 0.0 \times 0 = 0.1}$
$\mathbf{since\ z \geq 0; \hat{y} = 1}$
$\mathbf{y \neq \hat{y}}$, Thus update: $\mathbf{0.1 + 0.1(0 - 1) = 0.1 - 0.1 = 0.0}$

Epoch 2 - Re-evaluating All Points

x_1	x_2	y	z	$\hat{y}$	**Update?**
0	0	0	0.1	1	Yes, $w_0 = 0.1 - 0.1 = 0.0$
0	1	1	0.0	1	?
1	0	1	0.2	1	?
1	1	1	0.2	1	?

Repeat for all inputs

3.1.5 Start of Epoch – 2.

Epoch 2 - Re-evaluating All Points

x_1	x_2	y	z	$\hat{y}$	**Update?**
0	0	0	0.1	1	Yes, $w_0 = 0.1 - 0.1 = 0.0$
0	1	1	0.0	1	No
1	0	1	0.2	1	No
1	1	1	0.2	1	No

- Final Learned Weights:

  1. $w_1 = 0.2$
  2. $w_2 = 0.0$
  3. $w_0 = 0.0$

At this point, all classifications are correct, and no weight updates occur. The perceptron has converged.

3.1.6 Final Decision Boundary.

Step 4: Final Decision Boundary

Final Equation:

$$z = 0.2x_1 + 0.0x_2 + 0.0$$

Classification Rules:

- If $x_1 = 1$ or $x_2 = 1$, then $z \geq 0 \Rightarrow y = 1$.
- If $x_1 = 0$ and $x_2 = 0$, then $z < 0 \Rightarrow y = 0$.

Conclusion: The perceptron successfully learns the OR function.

3.2 Python Implementations.

Class Definition:

```
import numpy as np
class Perceptron:
    """
    A simple Perceptron classifier for binary classification.
    Attributes:
    -----------
    weights : np.ndarray
        Array of weights including the bias term, initialized randomly.
    learning_rate : float
        The step size for weight updates.
    epochs : int
        The number of training iterations over the dataset.

    Methods:
    --------
    step_function(z)
        Activation function that returns 1 if z >= 0, else 0.

    predict(x)
        Predicts the output for a given input sample x.

    train(X, Y)
        Trains the perceptron on the given dataset (X: inputs, Y: target labels).
    Example:
    --------
    X = np.array([[0, 0], [0, 1], [1, 0], [1, 1]])
    Y = np.array([0, 1, 1, 1]) # OR function
    perceptron = Perceptron(input_size=2, learning_rate=0.1, epochs=10)
    perceptron.train(X, Y)
    print(perceptron.predict([1, 1])) # Expected output: 1
    """
```

3.2 Python Implementations.

Defining " init" for initializations :

```
def __init__(self, input_size, learning_rate=0.1, epochs=10):
        """
        Initializes the perceptron with small random weights.
        Parameters:
        -----------
        input_size : int
            Number of input features.
        learning_rate : float, optional (default=0.1)
            Step size for weight updates.
        epochs : int, optional (default=10)
            Number of times to iterate over the dataset.
        """
        self.weights = np.random.rand(input_size + 1) * 0.2 - 0.1 # Small random weights
        self.learning_rate = learning_rate
        self.epochs = epochs
```

3.2 Python Implementations.

Making Decision using Step Function.

```
def step_function(self, z):
        """
        Step activation function.

        Parameters:
        -----------
        z : float
            Weighted sum of inputs and weights.

        Returns:
        --------
        int
            1 if z >= 0, else 0.
        """
        return 1 if z >= 0 else 0
```

3.2 Python Implementations.

Making a Prediction Function:

```python
def predict(self, x):
    """
    Predicts the output for a given input sample.

    Parameters:
    -----------
    x : array-like
        Input feature vector.

    Returns:
    --------
    int
        Predicted class label (0 or 1).
    """
    x = np.insert(x, 0, 1) # Bias term
    z = np.dot(self.weights, x)
    return self.step_function(z)
```

3.2 Python Implementations.

Making a Prediction Function:

```python
def predict(self, x):
        """
        Predicts the output for a given input sample.

        Parameters:
        -----------
        x : array-like
            Input feature vector.

        Returns:
        --------
        int
            Predicted class label (0 or 1).
        """
        x = np.insert(x, 0, 1) # Bias term
        z = np.dot(self.weights, x)
        return self.step_function(z)
```

3.2 Python Implementations.

Implementing Perceptron Learning Algorithm for Training:

```
def train(self, X, Y):
        """
        Trains the perceptron using the perceptron learning rule.

        Parameters:
        -----------
        X : np.ndarray
            Training data (each row is an input sample).
        Y : np.ndarray
            Corresponding target labels (0 or 1).

        Prints:
        -------
        Updates the weights and prints them after each epoch.
        """
        X = np.c_[np.ones(X.shape[0]), X] # Add bias term to input
        for epoch in range(self.epochs):
            for i in range(X.shape[0]):
                z = np.dot(self.weights, X[i])
                y_pred = self.step_function(z)
                error = Y[i] - y_pred
                self.weights += self.learning_rate * error * X[i]
            print(f"Epoch {epoch+1}, Weights: {self.weights}")
```

3.2 Python Implementations.

Putting it all Together:

```python
import numpy as np
# I Removed Docstring to Fit in the Slide # Not a Good Practise.
class Perceptron:
    def __init__(self, input_size, learning_rate=0.1, epochs=10):
        self.weights = np.random.rand(input_size + 1) * 0.2 - 0.1 # Small random weights
        self.learning_rate = learning_rate
        self.epochs = epochs

    def step_function(self, z):
        return 1 if z >= 0 else 0

    def predict(self, x):
        x = np.insert(x, 0, 1) # Bias term
        z = np.dot(self.weights, x)
        return self.step_function(z)

    def train(self, X, Y):
        X = np.c_[np.ones(X.shape[0]), X] # Add bias term to input
        for epoch in range(self.epochs):
            for i in range(X.shape[0]):
                z = np.dot(self.weights, X[i])
                y_pred = self.step_function(z)
                error = Y[i] - y_pred
                self.weights += self.learning_rate * error * X[i]
            print(f"Epoch {epoch+1}, Weights: {self.weights}")
```

3.2 Python Implementations.

Testing:

```python
# Example usage with adjustable input size
learning_rate = 0.1
epochs = 10
input_size = 3
# Create training data (modify as needed for different functions and input sizes)
X = np.array([
    [0, 0, 0],
    [0, 0, 1],
    [0, 1, 0],
    [0, 1, 1],
    [1, 0, 0],
    [1, 0, 1],
    [1, 1, 0],
    [1, 1, 1],
])
# Example: OR function with 3 inputs
Y = np.array([0, 1, 1, 1, 1, 1, 1, 1])
# Initialize and train the perceptron
perceptron = Perceptron(input_size, learning_rate=learning_rate, epochs=epochs)
perceptron.train(X, Y)
# Test the perceptron
test_samples = X # Use the same training samples for testing
print("\nPredictions:")
for sample in test_samples:
    print(f"Input: {sample}, Prediction: {perceptron.predict(sample)}")
```

3.3 Let's Try for XOR.

XOR Problem Recap

XOR Problem Recap:
The XOR (exclusive OR) gate produces the following outputs:

x_1	x_2	Output (Y)
0	0	0
0	1	1
1	0	1
1	1	0

We need to train a perceptron to solve this. The perceptron computes the weighted sum of inputs, applies a step function, and updates the weights based on the error.

3.3.2 Step – 1 – Initialization.

Step 1: Initialize Weights and Parameters

Start by initializing the weights w_0, w_1, w_2 (bias and input weights) to small random values (e.g., 0.2, -0.1). Set the learning rate η to 0.1.

$$w_0 = 0.1, \quad w_1 = 0.2, \quad w_2 = -0.1, \quad \eta = 0.1$$

3.3.3 Step – 2 – Weighted Sum and Step Function.

Step 2: Compute Weighted Sum and Apply Step Function

For each input pair (x_1, x_2), compute the weighted sum $z = w_1x_1 + w_2x_2 + w_0$, then apply the step function:

$$\hat{y} = \begin{cases} 1 & \text{if } z \geq 0 \\ 0 & \text{if } z < 0 \end{cases}$$

If the prediction $\hat{y}$ is incorrect, update the weights:

$$w_i = w_i + \eta(y - \hat{y})x_i \quad \text{for } i = 1, 2$$

$$w_0 = w_0 + \eta(y - \hat{y})$$

3.3.4 Step – 3 – Training for XOR.

Step 3: Training for XOR

Let's train using the XOR truth table.

Epoch 1 - Update for Each Training Example:

Initial weights: $w_0 = 0.1, w_1 = 0.2, w_2 = -0.1$

Training Example 1: (0, 0) $\rightarrow$ Expected $y = 0$

- Compute weighted sum:

$$z = (0.2 \times 0) + (-0.1 \times 0) + 0.1 = 0.1$$

- Apply step function:

$$\hat{y} = 1 \quad (\text{since } 0.1 \geq 0)$$

- Misclassified, update weights:

$$w_1 = 0.2 + 0.1 \times (0 - 1) \times 0 = 0.2 \quad (\text{no change})$$

$$w_2 = -0.1 + 0.1 \times (0 - 1) \times 0 = -0.1 \quad (\text{no change})$$

$$w_0 = 0.1 + 0.1 \times (0 - 1) = 0.0$$

3.3.4 Step – 3 – Training for XOR.

Step 4: Training Example 2

Training Example 2: (0, 1) to Expected $y = 1$

- Compute weighted sum:

$$z = (0.2 \times 0) + (-0.1 \times 1) + 0.0 = -0.1$$

- Apply step function:

$$\hat{y} = 0 \quad \text{(since -0.1 ¡ 0)}$$

- Misclassified, update weights:

$$w_1 = 0.2 + 0.1 \times (1 - 0) \times 0 = 0.2 \quad \text{(no change)}$$

$$w_2 = -0.1 + 0.1 \times (1 - 0) \times 1 = 0.0$$

$$w_0 = 0.0 + 0.1 \times (1 - 0) = 0.1$$

3.3.4 Step – 3 – Training for XOR.

Step 5: Training Example 3

Training Example 3: (1, 0) → Expected $y = 1$

- Compute weighted sum:

$$z = (0.2 \times 1) + (0.0 \times 0) + 0.1 = 0.3$$

- Apply step function:

$$\hat{y} = 1 \quad (\text{since } 0.3 \geq 0)$$

- Correctly classified, no update.

3.3.4 Step – 3 – Training for XOR.

Step 6: Training Example 4:

Training Example 4: (1, 1) → Expected $y = 0$

- Compute weighted sum:

$$z = (0.2 \times 1) + (0.0 \times 1) + 0.1 = 0.3$$

- Apply step function:

$$\hat{y} = 1 \quad (\text{since } 0.3 \geq 0)$$

- Misclassified, update weights:

$$w_1 = 0.2 + 0.1 \times (0 - 1) \times 1 = 0.1$$
$$w_2 = 0.0 + 0.1 \times (0 - 1) \times 1 = -0.1$$
$$w_0 = 0.1 + 0.1 \times (0 - 1) = 0.0$$

3.3.5 Can you try for Epoch – 2.

What happened?

3.3.6 Realization of "XOR" Problem.

Step 7: Epoch 2 and Final Conclusion

Epoch 2

Repeat the process for another epoch (i.e., another pass through the training examples). Students will realize that the perceptron does not converge and continues making mistakes on the XOR problem.

Final Conclusion: After Epoch 2, if the perceptron still doesn't learn the XOR function, it demonstrates that XOR is not linearly separable. A single-layer perceptron cannot solve the XOR problem. The perceptron will keep making errors because XOR needs a more complex decision boundary.

3.4 Python Implementation.

- Check the Starter Code:

4. The Perceptron for "XOR".

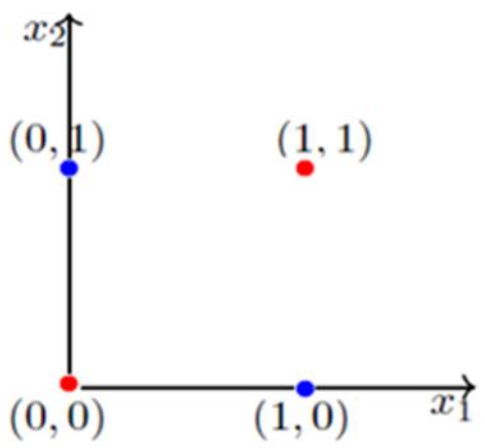
{Understanding Minsky and Papert Correction}

4.1 Why the perceptron will not converge for XOR?

- XOR is not linearly separable:

x_1	x_2	XOR	
0	0	0	$w_0 + \sum_{i=1}^{2} w_i x_i < 0$
1	0	1	$w_0 + \sum_{i=1}^{2} w_i x_i \geq 0$
0	1	1	$w_0 + \sum_{i=1}^{2} w_i x_i \geq 0$
1	1	0	$w_0 + \sum_{i=1}^{2} w_i x_i < 0$

$$w_0 + w_1 \cdot 0 + w_2 \cdot 0 < 0 \implies w_0 < 0$$
$$w_0 + w_1 \cdot 0 + w_2 \cdot 1 \geq 0 \implies w_2 > -w_0$$
$$w_0 + w_1 \cdot 1 + w_2 \cdot 0 \geq 0 \implies w_1 > -w_0$$
$$w_0 + w_1 \cdot 1 + w_2 \cdot 1 \geq 0 \implies w_1 + w_2 < -w_0$$

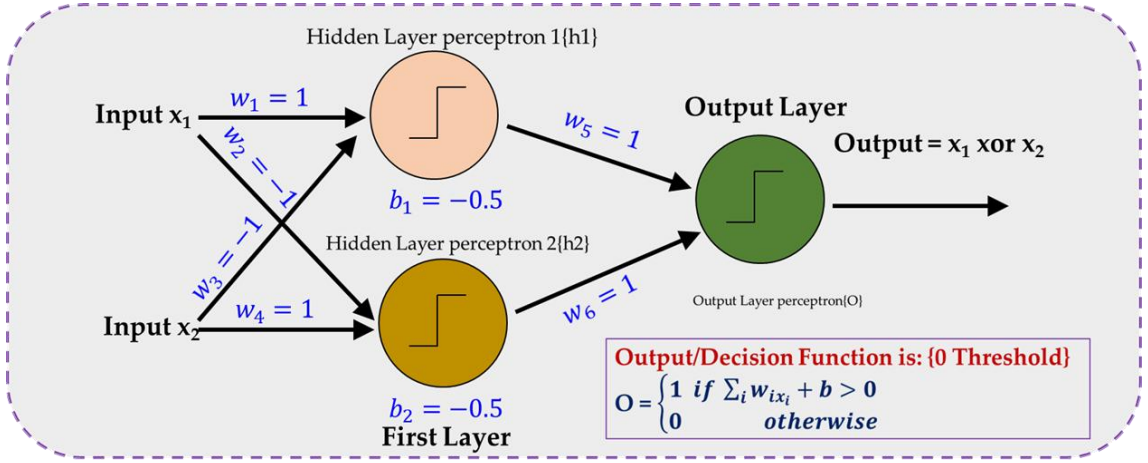


4.2 Why the perceptron will not converge for XOR?

- **The perceptron update rule keeps adjusting the weights:**
  - After each epoch, if the perceptron makes a mistake
    - (i.e., when predicted output $\hat{y}$ does not match the expected output y), it updates the weights using the rule:
      - $\mathbf{w_i = w_i + \eta(y - \hat{y})x_i}$
  - Since XOR is not linearly separable, the perceptron will continue to misclassify certain points.
  - As long as the perceptron misclassifies an example,
    - it will keep adjusting the weights based on the learning rule.
  - When will it stop?
- **Probably Never – The weights are constantly updated:**
  - In the case of XOR, the perceptron will continuously misclassify some examples because it cannot find a line (or hyperplane) that divides the data correctly.
  - For each mistake, the weights will be adjusted, but there will always be some examples where the perceptron gets it wrong, causing further weight updates.
  - Thus, No Convergence.

4.3 Solving the "XOR" with Multilayer Perceptron.

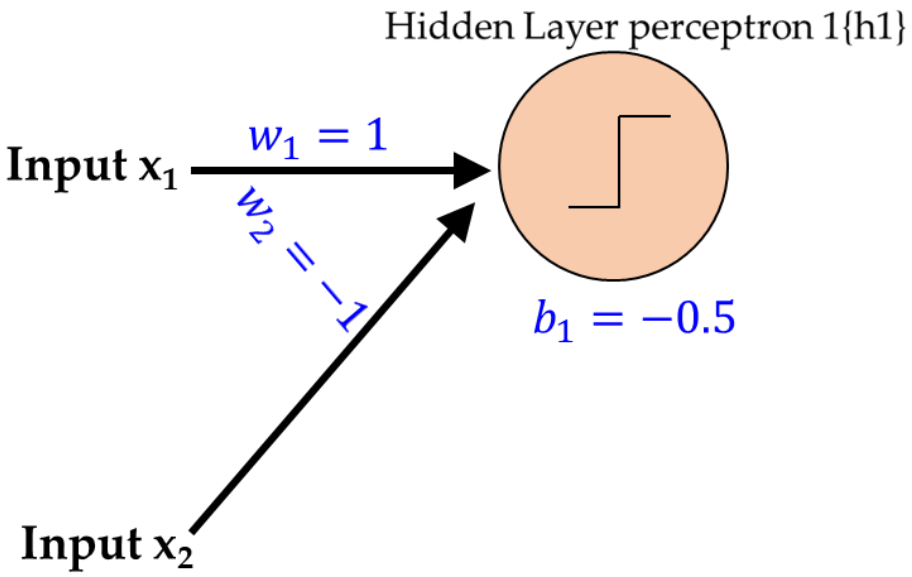
- Below Perceptron can solve for "XOR":



x_1	x_2	x_1 xor x_2
0	0	0
0	1	1
1	0	1
1	1	0

Solving for First Input Row @h1

x_1	x_2	x_1 xor x_2
0	0	0



Output of h1O1

$$h1O1 = \sum_i w_i x_i + b > 0$$
$$= 1 \times 0 + (-1) \times 0 + (-0.5) > 0$$
$$= 0 + 0 - 0.5 > 0 = -0.5 > 0$$

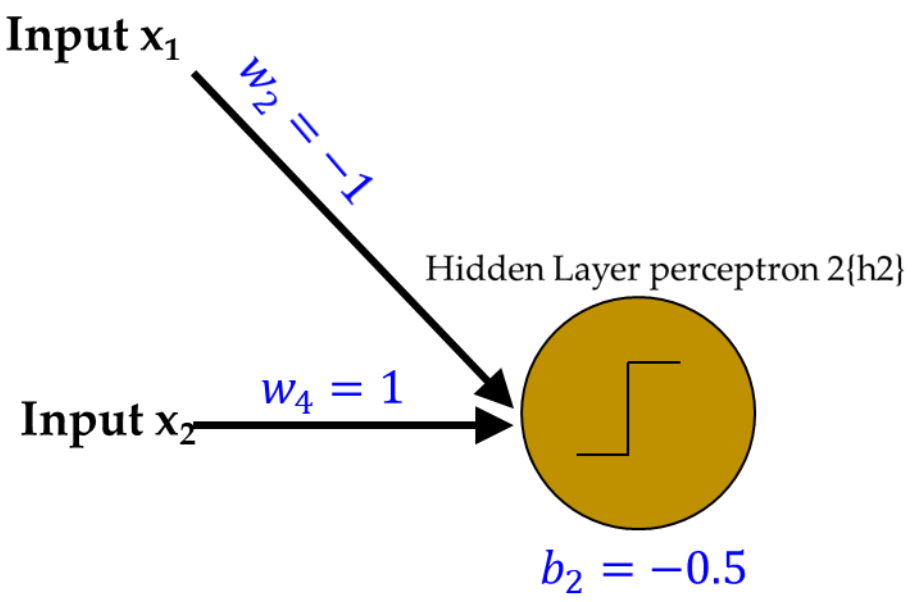
False Thus assigned class: h1o1 = 0

Solving for First Input Row @h2

x_1	x_2	x_1 xor x_2
0	0	0

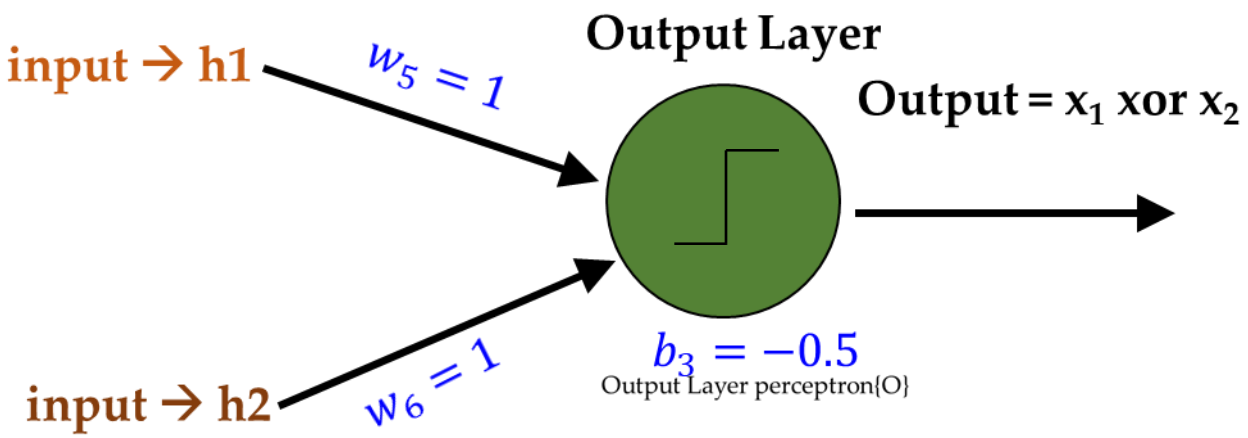
Your Turn to solve.

?



Solving for First Input Row @O

x_1	x_2	x_1 xor x_2
0	0	0



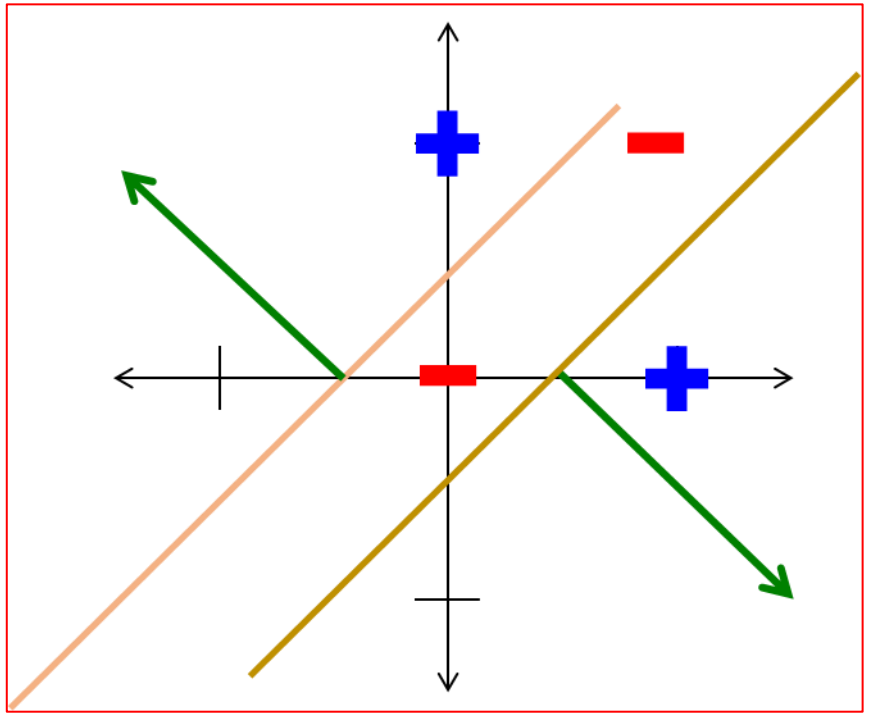
Your Turn to solve.

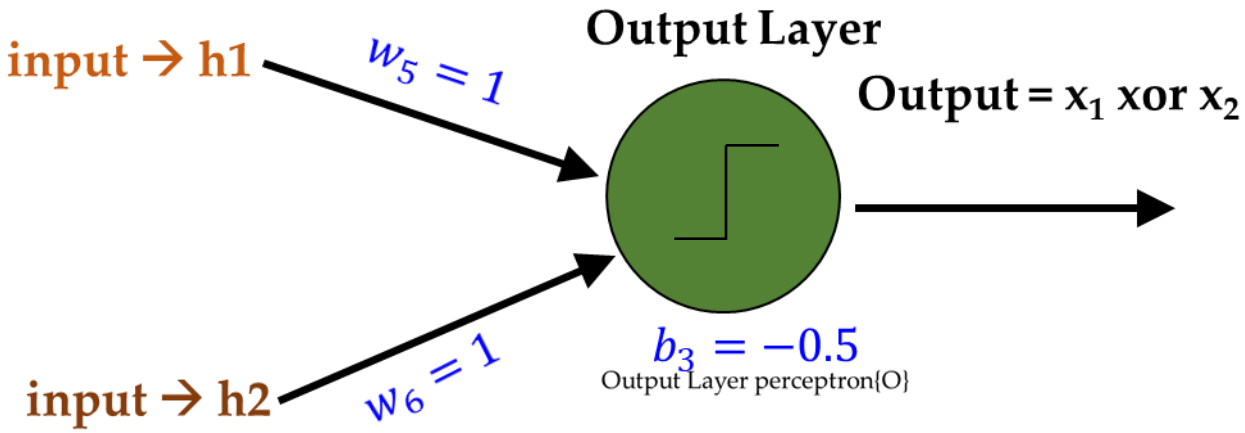
?

Did it Work? If Yes what type of Decision Boundary it Created?

Solving for First Input Row @O

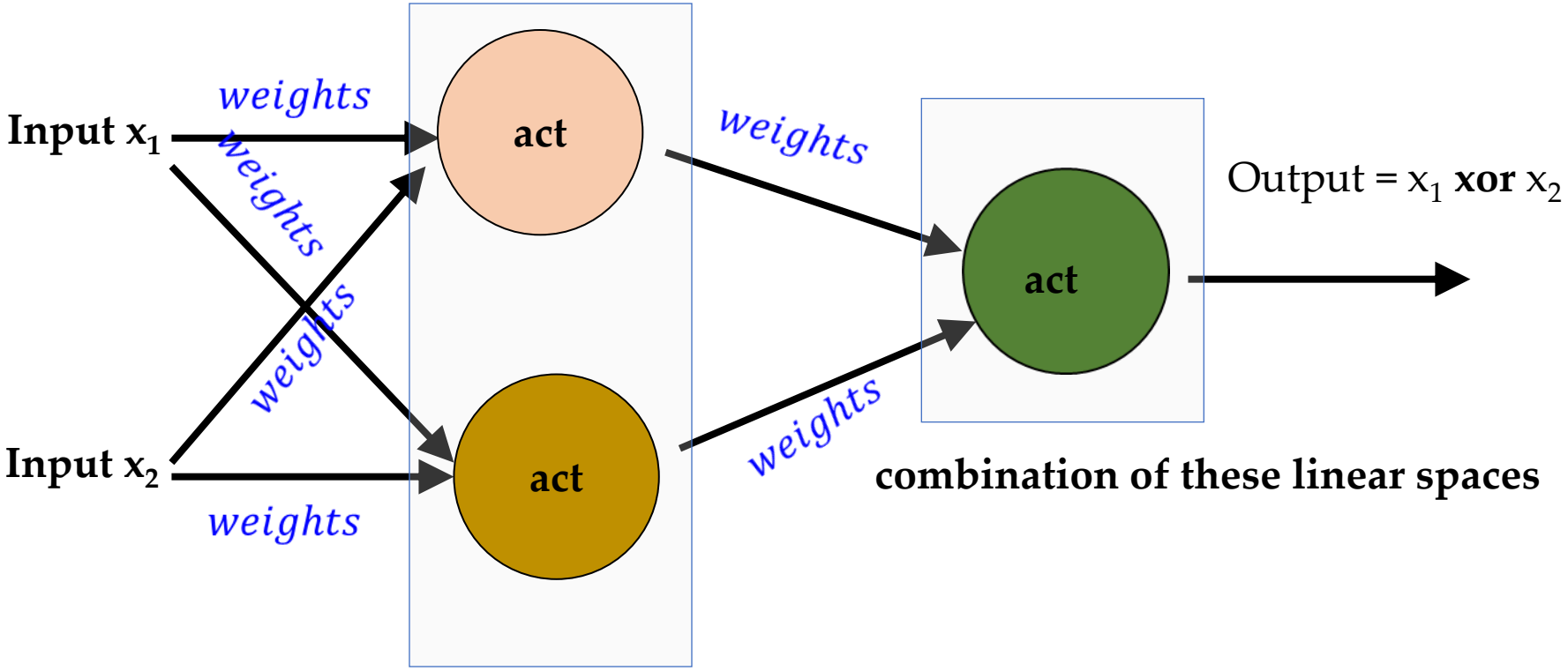
x_1	x_2	x_1 xor x_2
0	0	0

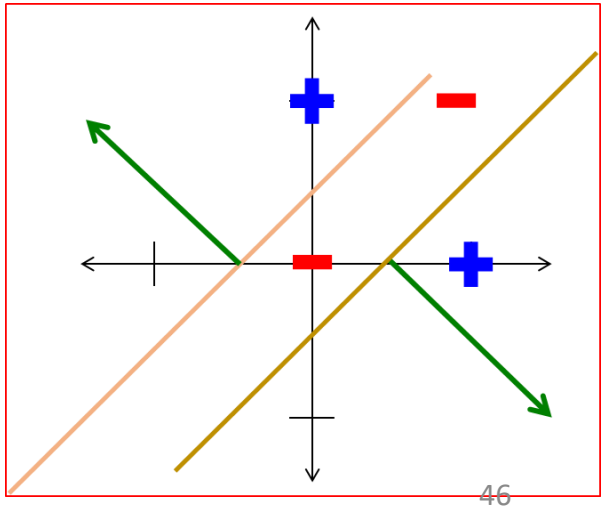




Can you solve for all the inputs. Also try for three inputs.

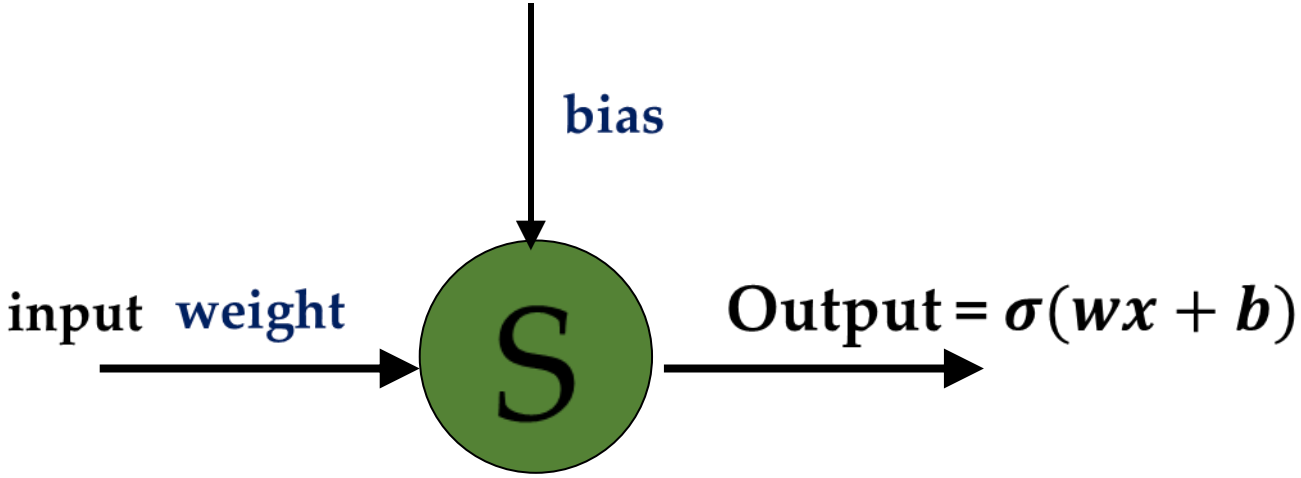
Decision Boundary for MLP:

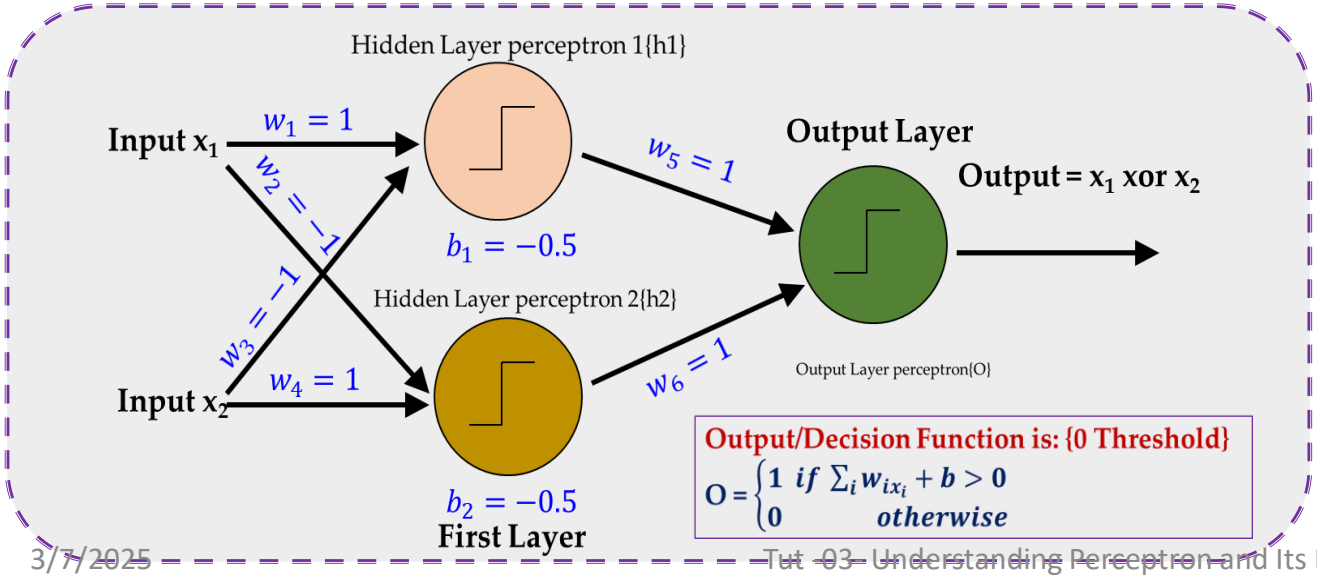




Home Work – Now Solve with Sigmoid Neuron.

x_1	x_2	x_1 xor x_2
0	0	0
0	1	1
1	0	1
1	1	0





The – End.